

Farzad Avishan
Operations research scientist
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Summary

Operations research and data scientist with over 3 years of experience in optimization and predictive modeling. My strong operational research background and proven machine learning skills enable me to design algorithms to solve complex problems and develop predictive models for large-scale data analysis. I have a track record of successfully implementing solutions that enhance decision-making and operational efficiency in various industries. I hold an open work permit, which is valid until February 2028.

LinkedIn: www.linkedin.com/in/farzad-avishan/

Website: www.farzadavishan.com/

Work experience

Air Canada, Canada, 2024 October- current (Research Scientist: Data& AI)

Main responsibilities

- Use optimization techniques to support the business
- Establish processes for large-scale data analyses, model development, validation, and implementation.
- Conduct exploratory data analysis to identify trends, patterns, and relationships in data.
- Collaborate with data professionals, such as data engineers and machine learning engineers, to ensure data pipeline and model integration.

HEC Montreal, Canada, 2023-2024, October (Post-doctoral research fellow)

Main responsibilities:

- Develop optimization models and solution approaches for various problems
- Conduct research in the field of specialization and publish findings in scholarly journals or books
- Provide professional consultative services to industry and private individuals

Ozyegin University, Istanbul, Turkey, 2021-2023 (Research assistant)

Main responsibilities:

- Design optimization models and solution approaches for diverse applications.
- Perform literature reviews and research activities to support academic studies.
- Compile and analyze research results and contribute to the preparation of scholarly publications.

Dadash Baradar Ind. Co. Tabriz, Iran, 2016-2017 (System analysis expert)

Main responsibilities:

- Develop predictive models to forecast demand on an annual, monthly, and weekly basis
- Manage production line setup project

Hadid Parlakh Co., Tabriz, Iran 2011-2015 (Quality control expert)

Main responsibilities:

- Utilize statistical control techniques to monitor and control the quality of production processes
- Develop dashboards to track quality metrics, identify trends, and highlight areas for improvement
- Identifying out-of-range processes and those needing calibration

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Consultant project

Clear D3, 2023

Project description: Utilizing historical data to develop a predictive model for forecasting transportation costs in upcoming periods

Key Achievements:

- Generated predictive models to approximate future transportation costs
- Enabled the company to use model outputs for more informed customer price negotiations
- Contributed to improved pricing strategies, enhancing the company's competitive edge in the market

Education

Ph.D. Industrial Engineering

Ozyegin University, Istanbul, Turkey, Received in March 2023

Research field: Applications of optimization under uncertainty methods on production planning, supply chain and logistics problems

Thesis Title: Applications of Robust Optimization in Logistics and Production Planning (CGPA: 3.87 out of 4)

M.Sc. Industrial Engineering

Mazandaran University of Science and Technology, Babol, Iran, Received in September 2014

Thesis Title: A New Optimization Model for Production and Distribution of Supply Chain with Green Logistic Approach

Published papers

All papers are available at: <https://scholar.google.ca/citations?user=WfsfoxwAAAAJ&hl=en>

Technical skills, interests, and expertise

Combinatorial optimization

Design and management of large-scale networks

Heuristic algorithms

Machine learning

Software skills

Programming Languages: Python, C++

Specialized Software: Gurobi, FICO Xpress, CPLEX, GAMS, MATLAB

Version Control: GitHub/ Git